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MIL-I-22560B(SHIPS)
1 December 1969
SUPERSEDING
MIL-I-22560A(SHIPS)
16 August 1963
(See 6.5)

MILITARY SPECIFICATION

INTERCOMMUNICATION STATIONS (HANDS-FREE OPERATED REPLY REPRODUCER) NAVAL SHIPBOARD

1. SCOPE

1.1 This specification covers LS-458/SIC "hands-free" operated intercommunication stations for use in Naval shipboard announcing systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-T-27 - Transformers and Inductors (Audio, Power and High Power Pulse), General Specification For.
MIL-C-62 - Capacitors, Fixed, Electrolytic (DC, Aluminum, Dry Electrolytic, Polarized), General Specification For.
MIL-C-915 - Cable, Electrical, Special Purpose, General Specification For.
MIL-I-983 - Interior Communication Equipment; Naval Shipboard, Basic Design Requirements For.
MIL-C-2194 - Cables, Power, and Control Reduced Diameter Types, (Shipboard Use).
MIL-C-3965 - Capacitors, Fixed, Electrolytic (Nonsolid Electrolytic) Tantalum, General Specification For.
MIL-L-12606 - Loudspeaker, Permanent Magnet (Unencased, 4-inch Diameter Cone, 2-Watt; Fungus-, Gunblast-, And Immersion-Resistant), Type LS-445/U.
MIL-M-15071 - Manuals, Technical Equipment and Systems Content Requirements For.
MIL-E-17362 - Electronic Repair Parts Requirements, Procedures For Provisioning Technical Documentation and Stock Numbering.
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts; Packaging and Packing of.
MIL-I-17928 - Intercommunication Stations (Naval Shipboard Use).

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-200 - Electron Tubes, Selection of.
MIL-STD-275 - Printed Wiring for Electronic Equipment.
MIL-STD-701 - Lists of Standard Semiconductor Devices.

DRAWINGS

MILITARY

9000-S6502-74117 - Receptacle Announcing Equipment Plug and Receptacle Sym. 723, 724, and 725.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Applications for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 First article sample. Prior to beginning production a sample shall be examined and tested as specified in 4.2 (see 6.3).

3.2 Description. The LS-458/SIC intercommunication station, hereinafter referred to as the "equipment", shall have selector switches for calling two stations, a busy light, one press to-talk switch, a microphone receptacle, one speaker-microphone switch, one remote-normal switch and one volume control switch. The equipment shall have a transistorized amplifier-speaker for voice amplification and transmission and shall have a signal lamp associated with each station selector switch. The equipment shall be designed for installation in protected locations.

3.3 General features. The equipment shall be in accordance with the following paragraphs of MIL-I-983 in addition to the requirements specified herein. (Whenever a requirement of MIL-I-983 conflicts with a requirement of this specification, the requirement of this specification shall govern.)

General requirements
Definitions
Materials, general (see 3.4)
Substitution of (equal or superior) materials or parts
Fungus-inert materials
Flammable materials
Arc-resistant materials
Toxic materials
Wood
Aluminum
Nonferrous material (except aluminum)
Zinc
Springs (material)
Other metals
Plastics
Ceramics
Impregnating, embedding and encapsulating compounds
Lubricants and lubrication
Painting
Protection against corrosion
Bolts, machine screws, studs and nuts
Gaskets
Locking devices
Washers
Parts-electrical-general
Electron tube or capacitor sockets
Capacitors
Variable resistors
Transformers and inductors (see 3.18)
Relays (see 3.19)
Electrical tapes
Batteries
Dial illumination lamps (see 3.20)
Switches
Indicator lights and lampholders
Fuses
Enclosure-degree of (see 3.9)
Enclosure-mounting
Minimum sheet metal thicknesses (watertight enclosures)
Cable entrance
Threaded devices
Rounded corners and edges

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Structural welding
Temperature and humidity
Shock, vibration and inclination (vital vibration test required)
Primary power supply circuits
Power supply tolerances (normal power supply, 115 volts, 60 cycle, single phase)
Shielding and radio frequency noise reduction
Ground potential and grounding
Soldering
Electrical parts mounting
Internal subassembly connection
Terminal boards and terminals
Wiring
Color coding
Electrical insulation
Dielectric strength and insulation resistance clearances
Drawings-general
Drawings-preliminary
Drawings-working
Schematic diagrams
Wiring diagram
Drawing list
Assembly drawings
Interchangeability and standardization (see 3.8)
Manuals (see 3.29)
Repair parts (electronic)
Designation and marking (see 3.27)
Item names and nomenclature
Workmanship

3.4 Materials. Materials used in the construction of the equipment shall be as specified herein or of an equivalent material when specifically approved by the command or agency concerned (see 3.3).

3.5 Design. The equipment shall be designed to provide naturalness and intelligibility of speech under quiet conditions, penetrating ability of speech under noisy conditions, dependability of operation, and accessibility for servicing.

3.6 Fuseholders. Fuseholders shall be of a spring clip type and shall be silver plated.

3.7 Switches. The handles or keys of rotary type selector switches shall be provided with mechanical stops to prevent undue strain on the switch handle and shaft, and to prevent rotation of the fixed portion of the switch. All switches shall, after 100,000 operations, when energized at their rated loads, be capable of performing both mechanical and electrical functions for which they are intended.

3.8 Interchangeability. The equipment shall conform to the following:

(a) The size and weight shall be not greater than the following:

Case	Overall (including mounting plate and handle grips)
Width 6-1/2 inches	6-1/2 inches
Depth 8 inches	9 inches
Height 6-1/2 inches	9-3/4 inches
Weight 25 pounds	

(b) The microphone receptacle shall be in accordance with Drawing 9000-S6502-74117. Wiring to this receptacle shall be as shown in MIL-I-17928.

3.9 Enclosure. A complete equipment enclosure shall consist of a case and removable front panel designed for mounting on a bulkhead and in a panel. The enclosure shall be steel or aluminum. The enclosure shall be dripproof in accordance with MIL-I-983.

3.10 Front panel and case securing bolts. Through bolting for the purpose of securing the front panel to the case and for securing the case in its mounted position will be permitted. All securing bolts shall have hexagonal heads with screwdriver slots. Sufficient space shall be allowed around bolt head surfaces to permit removal with a socket wrench.

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3.11 Front panel surface. The front panel shall be free of unnecessary projections. The depth of all operating controls in the nonoperated position shall be approximately the same distance from the front panel surface.

3.12 Handle grips. Handle grips shall be provided on the front panel to facilitate withdrawal of the front panel from the case.

3.13 Electrical connections.

3.13.1 Ship's wiring connections. A terminal board shall be provided for connecting ship's wiring to the equipment. The terminal board location shall be such that ship's wiring can be connected to the terminal board without removing the equipment from its case. Sufficient terminals shall be provided on the terminal board to allow ship's wiring connections for all modes of operation.

3.13.2 Plug-in connectors. Multi-contact plug and jack connectors shall be provided which will disconnect the unit from ship's wiring when it is removed from its case. The plug shall be mounted on the equipment and the jack shall be mounted in the rear of the case such that the plug engages in the jack when the equipment is placed in its case. The connectors shall employ a floating type of mount to assure positive alignment and positive connection.

3.13.2.1 Alignment guides. Alignment guides shall be provided to assure positive alignment between the plug and jack connectors. The guides shall support the equipment during insertion and removal intervals while in the case. The guides shall prevent any damage to the plug-in connectors or any part of the equipment during insertion and removal of the unit from its case.

3.14 Drainage. Drainage shall be provided for any panel recesses which otherwise would be susceptible to collection of water.

3.15 Control knobs. The size and shape of the operating control knobs shall be approved by the procuring activity for the function for which they are intended.

3.15.1 The position of all switch controls shall be clearly visible in the normal or operated position.

3.15.2 Knobs and shafts shall have flats or rectangular shapes on the connecting parts and shall be secured with a through roll-pin or other means approved by the procuring activity.

3.16 Gaskets. Gaskets shall be used, if necessary, for making dripproof seals between the case and front panel assembly. Gaskets shall be used for operating controls of the translating and rotary motion types.

3.17 Servicing. With the equipment removed from its case, all components shall be readily accessible for ease of replacement. When the equipment is removed from its case for servicing, all component parts shall be protected from possible damage when placed on any of its sides.

3.18 Transformers. Transformers shall be grade 5, class S, life expectancy X, as specified in MIL-T-27. Where special shielding problems exist, such as in input circuits of audio amplifiers, the use of encased type transformers employing high permeability alloy shields will be permitted.

3.19 Relays. Relays, if used, shall conform to MIL-I-983 and shall have removable dustproof covers.

3.20 Illumination. All illumination shall be red in accordance with MIL-I-983. Illumination lamps shall be located to permit direct replacement of each lamp individually from the exterior of the watertight enclosure.

3.21 Signal indicator lamps. The signal indicator lamps shall be individually located adjacent to the associated station selector switches. Signal indicator lamps shall be subminiature, omnidirectional, red plastic dome using a type T-1-3/4 lamp.

3.22 Station identification. This shall be a durable, abrasive-resistant laminated plastic strip suitable for engraving. The laminations shall be opaque black and translucent white. Guide lines shall be engraved on the marker to indicate the limits of the area to be engraved with the station designation. The engraved areas shall be at a sufficient depth to expose the translucent white lamination. The marker shall be a single strip removable from the front of the equipment.

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3.23 Rectifiers. Silicon type rectifiers shall be used to supply high voltage for the amplifier. A resistance shall be inserted in the high voltage circuit as a surge current limiter. Rectifiers shall be selected from approved types listed in MIL-STD-200.

3.24 Operational features. The operational features of the system shall be as specified in 3.24.1 to 3.24.10, inclusive. (Master stations referred to are type LS-519/SIC or LS-518/SIC units manufactured in accordance with MIL-I-17928).

3.24.1 LS-458/SIC calling master station. Operate station selector switch. If station selected is not busy (indicated by busy lamp) operate talk-listen switch and talk into speaker.

3.24.2 Busy master station. If the master station is busy, the busy indicator lamp will light. This is an indication that the master should not be called.

3.24.3 Operation of equipment with remote reproducer. This system shall consist of one LS-458/SIC station, one remote reproducer with one normally closed single pole switch. Circuits shall be provided in the equipment for the following operations:

(a) Operation of remote reproducer switch will connect remote reproducer to the input of the equipment amplifier and connect the equipment reproducer to the output of the equipment amplifier. The remote reproducer can then transmit speech.
(b) Operation of station selector switch on the equipment will connect the remote reproducer to input of the equipment amplifier and connect the equipment reproducer to output of the equipment amplifier. Speech can then be transmitted from the remote reproducer without operation of the remote reproducer switch.
(c) With equipment selector switch operated, this station can transmit speech to remote reproducer by operation of press-to-talk switch. Remote reproducer can reply with hands-free operation when the equipment "press to talk" switch is in restored position.

3.24.4 Master station calling equipment station. The equipment shall be capable of operating as follows:

(a) Normal operation to answer a call. Press talk-listen switch and talk into speaker. (However, even though unnecessary, if the conventional method of selecting the calling station by pressing the station selector switch is used, a busy light at the answering station shall light, but this shall not prevent the answering station from talking back to the calling station.)
(b) Hands free operation. With "Remote-Normal" switch on LS-458/SIC station in "Remote" position, operation of station selector switch on master station shall put the equipment in "hands-free talk" position. Operation of press-to-talk switch on master station shall cause equipment station to be in "Listen" position. When master station depresses release switch both stations shall go to standby condition.

3.24.5 Restoring station selector switch to normal shall also turn off busy light if selected station is busy, and restore the station to the normal receiving position.

3.24.6 Operation of a station selector switch shall cause the selected station (if not already busy) to become busy to all other stations in a system. Operation of this switch shall also cause the selecting station to become busy to all other stations in the system. Operation of a station selector switch shall energize the call light at the station selected provided the station is not busy.

3.24.7 Once communication has been established by operation of station selector switches, no other nonselected equipment shall be able to listen or to break in and talk to any units engaged in conversation.

3.24.8 Operation with handset or microphone shall be identical to the normal operation, except that talk-listen functions will be performed remotely by the press-to-talk switch in the handset or microphone.

3.24.9 Operation under busy condition. If, on selection of either station selector switch a busy indication is received, the switch shall be released immediately. This will place the unit in condition for the reception of an incoming call. When a station has received a busy indication, it shall not be possible to hear speech from the busy station.

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3.24.10 Parallel operation of units. When two equipments are connected in parallel operation, they shall be the same as for normal single operation with the following exceptions:

(a) Incoming speech shall be heard at both equipments.
(b) Either station may answer a call but not both. (Inadvertent answering by both units simultaneously will not cut off transmission of speech to the calling equipment.)
(c) Operation of a station selector at either of the paralleled equipments in addition to lighting the call light at the called station shall cause the call lamp in the unit to which it is paralleled to light.

3.25 Electrical features.

3.25.1 Circuit design. The circuit design shall perform the functions specified in 3.24 through 3.24.10.

3.25.2 Terminal markings. Terminal markings shall be in accordance with MIL-I-17928.

3.25.3 Interconnecting cables. Type TTHFWA cable conforming to MIL-C-915 will be employed for interconnection of audio and control circuits when installed. Reduced diameter cable conforming to MIL-C-2194 shall be employed for power and signal circuit common. A maximum length of 1500 feet shall be used for calculating line impedance of audio and signal circuits.

3.25.4 Volume control. The volume control shall provide a maximum attenuation of 30 ± 1 decibels (db) in five steps of 6 ± 1 db. No "OFF" position shall be incorporated in the control.

3.25.5 Press-to-talk switch. The press-to-talk switch shall be a nonlocking spring return switch. The nonoperated position of the switch shall be the normal incoming call position.

3.25.6 Speaker-microphone switch. The speaker-microphone switch shall be a DPDT toggle switch.

3.25.7 Speaker remote-normal switch. The speaker remote-normal switch shall be a SPST toggle switch.

3.25.8 Microphone and telephone handset. A connector shall be provided on all equipment. The purpose of the connector is to permit the use of either a microphone or telephone handset. The contractor of the intercommunication stations shall be responsible for the proper operation of the intercommunication stations when used with a type M-132/SIC microphone or an IC/MPH-2 handset. Facilities shall be provided to permit the use of a separate handset or microphone of the types specified herein. The contractor shall include in drawings and manuals for the intercommunication stations, the circuits, cable size and type, and electrical and acoustical characteristics of the microphone and telephone handset based upon information to be furnished by the command or agency concerned.

3.25.9 Loudspeakers. Loudspeakers shall be type LS-445/U in accordance with MIL-L-12606 and shall be identical for all stations.

3.25.10 Transformer polarities. Identical instantaneous alternating current (a.c.) polarities for all transformers shall be obtained at physically identical transformer terminals on all stations, when all stations are energized from the same source.

3.26 Amplifier.

3.26.1 Transistors and metallic rectifiers. Transistors shall be used in the design of all amplifiers. Transistors shall be types chosen from approved types listed in MIL-STD-701. Power rectifiers shall be chosen from types listed in MIL-STD-200.

3.26.2 Printed wiring. Printed wiring if used, shall be in accordance with MIL-STD-275 and type C lead terminations shall be employed.

3.26.3 Protection of signal voltage winding of power transformer. Fuses shall be incorporated in the signal voltage circuit to prevent damage to the windings of the transformer due to defective signal voltage circuits.

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3.26.3.1 Fuses shall be inserted in each lead of the primary power supply in the unit.

3.26.4 Capacitors. Tantalum electrolytic capacitors conforming to MIL-C-3965 shall be used in the amplifier assemblies.

3.26.4.1 Electrolytic capacitors in accordance with MIL-C-62 shall be used in power supply filter circuits only.

3.26.5 Personnel protection. In the case of voltages in excess of 240 volts, a red warning plate shall be mounted in a conspicuous location in accordance with MIL-I-983.

3.26.6 Heat dissipation. Components which generate heat shall be mounted to allow maximum heat dissipation through the external case.

3.26.7 Rated power output. Rated power output shall be 3 watts at a maximum distortion of 10 percent.

3.26.8 Output voltage. The output voltage when delivering rated power shall be 70 volts ± 4 volts root mean square (r.m.s.).

3.26.9 Amplifier gain. The amplifier gain shall be sufficient to deliver rated output power at rated output voltage with a total harmonic distortion of 10 percent with a 1000 cycle sine wave voltage equivalent in value to the voltage generated by the loudspeaker, when used as a microphone under the test conditions specified in 4.6.2 applied to the input terminals of the amplifier. The amplifier gain shall be such that the amplifier will also deliver rated output power at rated output voltage when tested with a type M-132/SIC microphone or equivalent which will be capable of delivering 8.7 millivolt (rv.) into a resistive load of 150 ohms.

3.26.10 Output voltage regulation. With a steady state sine wave input voltage, of the same value used to determine amplifier gain, applied to the input terminals of the amplifier, a variation of the amplifier output load from full load to no load shall not cause the output voltage to exceed 99 volts r.m.s. at any frequency between 200 and 8000 cycles per second (c.p.s.).

3.26.11 Stability. The amplifier shall show no evidence of instability or oscillation under any condition of operation.

3.26.12 Amplifier noise and microphonics. The amplifier noise and microphonics shall be not greater than the following values when tested as specified in 4.6.7.

(a) Component stationary. Maximum measured voltage shall be not greater than 0.22 volts r.m.s. (50 db below rated output voltage of 70 volts).
(b) Component vibrating. Maximum measured voltage shall be not greater than 0.31 volts r.m.s. (47 db below rated output voltage of 70 volts).

3.26.13 Overall frequency response. The overall frequency response using the loudspeaker as a microphone shall fall within the limits of the overall frequency response curves shown on figure 1 when tested as specified in 4.6.4 and 4.6.5.

3.26.14 Minimum sound pressure. The loudspeaker, when mounted in the front panel, shall be capable of producing the following minimum sound pressure at a distance of 2 feet in front of and on axis with the loudspeaker when measured as specified in 4.6.3.

Warble band c.p.s.	Output pressure db 1/
200 to 320	85
320 to 500	95
500 to 800	99
800 to 1250	99
1250 to 2000	99
2000 to 3200	99
3200 to 5000	99

1/ Above 0.000200 dynes per cm^2.

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3.26.15 Minimum off-axis sound pressure. The loudspeaker, when mounted in the front panel, shall be capable of producing the following minimum 45-degree off-axis sound pressure at a distance of 2 feet when measured as specified in 4.6.3.

Warble band c.p.s.	200 to 300	300 to 500
Output pressure db 1/	81	91
	500 to 800	800 to 1250
	96	96
	1250 to 2000	2000 to 3200
	96	96
	3200 to 5000	
	91	

1/ Above 0.000200 dynes per cm^2.

3.26.16 Amplifier endurance. After being tested as specified in 4.6.6, the equipment shall be capable of normal operation in every respect. There shall be no visible damage to any electrical or mechanical component of the equipment.

3.26.17 Signalling circuits. The busy, and call signals shall not fail to indicate nor shall speech transmission be audible at a station receiving a busy signal when tested as specified in 4.6.8.

3.26.18 Amplifier output circuits. The paralleling of amplifier output circuits which can occur if two stations call a third station simultaneously shall not cause a loss of signal greater than 2 db, nor shall the distortion exceed the maximum allowable distortion.

3.27 Designation and marking. An instruction plate containing abridged operating instructions shall be provided on the outside of the enclosure. Schematic circuit diagrams shall be provided and mounted inside the enclosure for use in servicing the equipment. The circuit diagram plates may be of aluminum or plastic material. Designation plates shall be provided for all control knobs. These plates shall be located adjacent to the knobs and shall clearly indicate the function and direction of rotation of the knobs.

3.28 Equipment repair parts. Equipment repair parts shall be in accordance with MIL-E-17362.

3.29 Manuals. The manual shall be type I in accordance with MIL-M-15071. The manual shall cover the equipment specified herein, and shall include microphone and handset operating techniques. The manual shall contain preliminary drawings, working drawings, and an illustrative schematic showing the interconnection of stations and a sufficient portion of the internal wiring of each station to enable a reader to determine the operation of the switches and relays and to service and install the equipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. First article inspection shall consist of the examination and tests specified in table I, which shall be performed in general in the order listed.

4.3 Sampling for quality conformance inspection.

4.3.1 Lot. All equipment presented for delivery at one time shall be considered a lot. The lot may include the entire contract quantity or it may be the production of any convenient time period.

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Table I - First article inspection.

Inspection	Requirement paragraph	Inspection reference
Examination	3.3	MIL-I-983
Operating	3.24	4.6.1
Supply line voltage and frequency variation	3.3	MIL-I-983
Shielding and radio frequency noise reduction	3.3	MIL-I-983
Temperature and humidity	3.3	MIL-I-983
Insulation resistance	3.3	MIL-I-983
Amplifier gain	3.26.9	4.6.2
Amplifier noise and microphonics	3.26.12	4.6.7
Overall frequency response	3.26.13	4.6.4 and 4.6.5
Minimum sound pressure	3.26.14	4.6.3
Minimum off-axis sound pressure	3.26.15	4.6.3
Amplifier endurance	3.26.16	4.6.6
Signalling circuits	3.26.17	4.6.8
Enclosure	3.3 and 3.9	MIL-I-983
Shock, vibration and inclination	3.3	MIL-I-983

4.3.2 Sampling for group A examination and tests. Sample equipments shall be selected in accordance with MIL-STD-105 at inspection level III for the examination and tests specified in 4.4. The acceptable quality level (AQL) shall be 2.5 percent.

4.3.3 Sampling for group B tests. Sample equipments shall be selected in accordance with inspection level S-3 of the table for small sample inspection of MIL-STD-105 for the tests specified in 4.4. The AQL shall be 4.0 percent.

4.3.4 Sampling for group C tests. Group C tests will be required by the command or agency concerned when the basic design of the equipment or the material of a vital part has been changed. One complete equipment shall be selected and subjected to the group C tests listed in 4.4.

4.4 Quality conformance inspection. The sample equipments selected in accordance with 4.3 shall be subjected to the examination and tests specified in table II. Tests shall be performed, in general, in the order listed.

Table II - Quality conformance inspection.

Inspection	Requirement paragraph	Inspection reference
Group A		
Examination	3.3	MIL-I-983
Operating	3.24	4.6.1
Amplifier gain	3.26.9	4.6.2
Group B		
Supply line voltage and frequency variation	3.3	MIL-I-983
Amplifier noise and microphonics	3.26.12	4.6.7
Overall frequency response	3.26.13	4.6.4 and 4.6.5
Minimum sound pressure	3.26.14	4.6.3
Minimum off-axis sound pressure	3.26.15	4.6.3
Signalling circuits	3.26.17	4.6.8
Group C		
Shielding and radio frequency noise reduction	3.3	MIL-I-983

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Table II - Quality conformance inspection. (Cont'd)

Inspection	Requirement paragraph	Inspection reference
Group C (Cont'd)		
Amplifier endurance	3.26.16	4.6.6
Temperature and humidity	3.3	MIL-I-983
Shock, vibration and inclination	3.3	MIL-I-983
Enclosure	3.3 and 3.9	MIL-I-983

4.5 Rejection. The results of the examination and each test shall be compared with specification requirements. Failure to conform to this specification for any group A, B or C examination and test shall be counted as a defect, and the equipment shall not be offered for delivery.

4.6 Test procedures.

4.6.1 Operating test. The operating test shall consist of a voice transmission test for each intercommunication station with a complete operational test using a type LS-519/SIC or LS-518/SIC unit.

4.6.2 Amplifier gain. The sound source for the test shall be the test laboratory speech generator (artificial voice) adjusted so as to deliver a sound pressure of 4 dynes per cm^2 at a frequency of 1000 c.p.s. under free field conditions at a point of axis and 18 inches in front of the ring on the speech generator. The loudspeaker to be tested shall be mounted in its front panel and placed 18 inches in front of and on axis of, and facing the ring on the speech generator. The loudspeaker shall be connected to a resistive load corresponding in value to the normal amplifier input impedance. The voltage developed across the resistor shall be measured to determine conformance with 3.26.9.

4.6.3 Sound output pressure test. Three intercommunication stations shall be connected together for normal operation. One of the stations shall be placed with its reproducer at a point 18 inches in front of and on axis and facing the ring of the artificial voice. A sound source from the artificial voice shall be adjusted so as to deliver a sound pressure of 4 dynes per cm^2 at the warble band frequencies specified in 3.26.14 under free field conditions at a point on axis and 18 inches in front of the ring on the artificial voice. The station under test shall have the selector key for the two other stations depressed and the press-to-talk switch shall be operated. The sound output pressures at the two other stations (with volume control in maximum position) shall be measured. Both stations being transmitted to shall be acoustically insulated from each other.

4.6.4 Overall frequency response test using loudspeaker as a microphone. Conditions of overall frequency response test shall be as specified in 4.6.3 except that, in lieu of the warble frequencies specified in 3.26.14, a constant sound pressure at a frequency which shall be varied continuously from 200 to 8000 c.p.s. shall be employed. The overall frequency response shall lie within the limits shown on figure 1.

4.6.5 Overall frequency response using a type M-132/SIC microphone. Test conditions shall be as specified in 4.6.4 except that the sound source pressure shall be 100 dynes per cm^2 and the distance from the microphone to the artificial mouth shall be 1/4 inch.

4.6.6 Amplifier endurance test. A resistive load of 1667 ohms shall be connected across the normal output terminals of a station. A 1000-cycle sine wave signal of such value as to produce rated output at rated output voltage shall be applied to the microphone input terminals. The station shall be energized and the press-to-talk switch shall be depressed for a period of 7 seconds and then released for a period of 15 seconds. Immediately upon completion of 20,000 operations the station shall be connected for normal operation. There shall be no noticeable change in the quality of speech transmitted, nor shall the output decrease greater than -2 db from the output measured at the beginning of the test. Amplifier endurance tests shall be conducted in the hands-free mode, at rated load, for a period of 24 hours.

4.6.7 Amplifier noise and microphonics. With a 150-ohm resistor connected to the microphone input terminals, and full power applied to the amplifier, the output voltage across a resistive load (of such value as to draw rated output power at 70 volts) shall be

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not greater than the following values when measured with an unweighted full range voltmeter. The measurement shall be made under two conditions of test, once while the component is held fixed and again while the component is under vibration in accordance with 3.26.12.

(a) Component stationary. Maximum measured voltage shall be 0.22 volt r.m.s. (50 db below rated output voltage of 70 volts).
(b) Component under vibration. Maximum measured voltage shall be 0.31 volt r.m.s. (47 db below rated output voltage of 70 volts).

4.6.8 Signalling circuits. A combination of three intercommunicating units shall be connected for normal operation. Cable used to connect the equipment shall be equivalent to the cables specified in 3.25.3. The equipment shall be tested in the following sequence:

(a) At station number 1, select station numbers 2 and 3. The call lights at station numbers 2 and 3 should light. Repeat this test ten times.
(b) At station Number 1, operate selector switch to select station Number 2. At station Number 3, operate selector switches to select stations Numbers 1 and 2, consecutively. The busy light at station Number 3 should light after each selection. This test shall be repeated ten times. A message shall be transmitted at station Number 1 during the fifth and tenth test cycles. A listening test shall be made at station Number 3 during the fifth and tenth test cycles. The transmission being made at station Number 1 shall not be audible at station Number 3.
(c) The test specified in 4.6.8(b) shall be repeated until each station in the system has served as the calling station (station Number 1) and as the busy testing station (station Number 3).
(d) Tests shall be conducted for compliance with 3.26.1d by operating station selector switches on both LS-458/SIC stations, simultaneously completing a call to a type LS-519/SIC or LS-518/SIC station.

4.7 Inspection of preparation for delivery. The packaging, packing and marking shall be inspected for compliance with section 5 of this document.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.4).

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts.

5.1.1 Intercommunication stations.

5.1.1.1 Preservation and packaging. Preservation and packaging which may be the supplier's commercial practice, shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing. Packing shall be accomplished in a manner which will insure acceptance by common carrier at lowest rate, and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice when such meets these requirements.

5.1.1.3 Marking. Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the suppliers commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.1.2 Onboard repair parts. Onboard repair parts shall be preserved and packaged by level A; packed by level C, and marked by levels A and C respectively in accordance with MIL-E-17555.

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5.2 Domestic shipment and storage or overseas shipment. The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 5.3 and 6.2).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made:

5.2.1.1 Preservation and packaging, packing and marking. The equipment and accessories, repair parts and technical publications shall be preserved and packaged by levels A or C; packed by levels A or B as specified, and marked in accordance with MIL-E-17555.)

5.3 Use of polystyrene (loose-fill) material.

5.3.1 For domestic shipment and early equipment installation and level C packaging and packing. Unless otherwise approved by the procuring activity (see 6.2), use of polystyrene (loose-fill) material for domestic shipment and early equipment installation and level C packaging and packing applications such as cushioning, filler and dunnage is prohibited. When approved, unit packages and containers (interior and exterior) shall be marked and labelled as follows:

"CAUTION

Contents cushioned etc. with polystyrene (loose-fill) material.
Not to be taken aboard ship.
Remove and discard loose-fill material before shipboard storage.
If required, recushion with cellulosic material bound fiber, fiberboard or transparent flexible cellular material."

5.3.2 For level A packaging and level A and B packing. Use of polystyrene (loose-fill) material is prohibited for level A packaging and level A and B packing applications such as cushioning, filler and dunnage.

6. NOTES

6.1 Intended use. The equipment covered by this specification is intended for intercommunication onboard Naval ships by means of voice amplification and station selection.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Preservation and packaging, packing and marking required if other than as specified in 5.1 (see 5.2).
(c) When polystyrene (loose-fill) material is approved (see 5.3).

6.3 First article. Invitations for bids should provide that the Government reserves the right to waive the requirement for the first article as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in Section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

Preparing activity:
Navy - SH
(Project 5835-N043)

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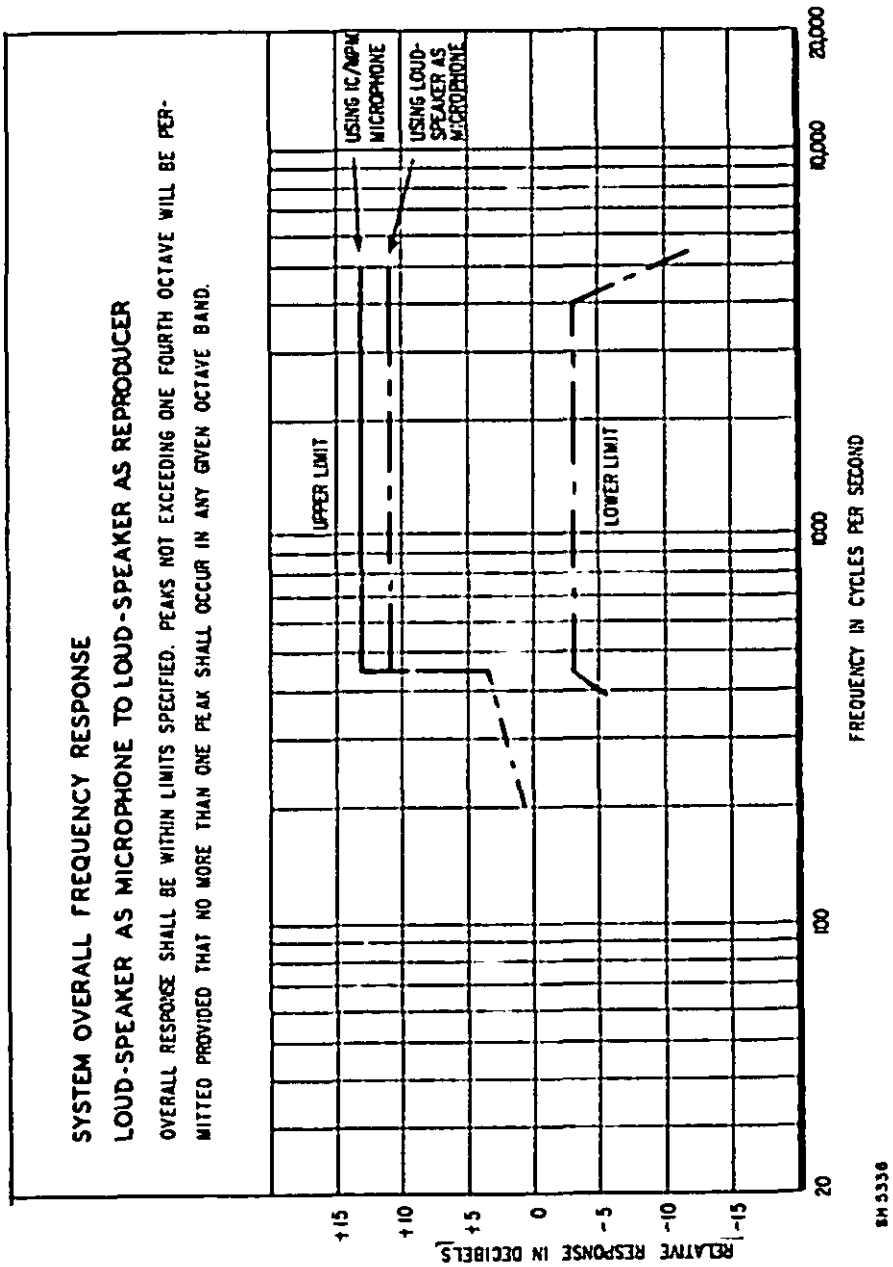


Figure 1 - System overall frequency response.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE 0101-807-2000

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